



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ3483-16 TEKNOLOGI WEB (WEB TECHNOLOGY)

Web Application & Robotic Process Automation (RPA)

**Created by: Marwan Sameh Raafat Elhennawy –
A22MJ3007**

1. Introduction

The DHL Knowledge Base Automation System is an integrated enterprise-style automation platform designed to transform raw operational data into structured knowledge base articles. It combines a web application, backend API services, Robotic Process Automation (RPA), and a Large Language Model (LLM) to automate the complete lifecycle of knowledge creation.

In many operational environments, knowledge management is still handled manually. Employees are required to take unstructured inputs such as emails, reports, scanned documents, and logs, and convert them into structured documentation. This manual approach is inefficient and often leads to inconsistent formatting, loss of important details, and delays in documentation. This project addresses these challenges by introducing an automated pipeline that performs this transformation without human intervention.

2. Problem Statement

Organizations such as logistics and warehouse operations generate large volumes of unstructured data on a daily basis. These include incident reports, operational logs, and communication records. When these inputs are manually processed into knowledge base systems, several issues arise. The process is time-consuming, it is prone to human error, and it results in inconsistent documentation standards. Additionally, important operational insights may be overlooked during manual summarization. As a result, there is a clear need for an automated system that can standardize and accelerate this transformation process.

3. Project Objectives

The primary objective of this system is to automate the conversion of unstructured operational data into structured knowledge articles. The system aims to achieve this by integrating multiple technologies into a unified workflow. It is designed to automate document ingestion using RPA, extract and process text from multiple file formats, and apply AI-based structuring using a large language model. The final output is stored in a structured database and presented through a web-based dashboard for user access and management.

4. System Overview

The DHL Knowledge Base Automation System is designed as an end-to-end automated pipeline that converts raw unstructured files into structured knowledge base articles. The process begins when files are uploaded either manually by users through the web interface or automatically through the UiPath automation workflow. These files are sent to the backend API, where an initial database record is created for tracking and processing.

Once received, the backend extracts text from the uploaded file based on its format, supporting document types such as PDF, DOCX, TXT, and image files using OCR where necessary. The extracted content is then cleaned and pre-processed to remove unnecessary formatting and prepare the data for analysis.

The cleaned text is sent to a locally hosted Large Language Model (LLM) through Ollama, where the AI restructures the unstructured content into a formatted knowledge base article. The generated output includes fields such as title, summary, steps, key points, risks, and notes. This structured article is then stored in the MySQL database and made available through the frontend dashboard for viewing and management.

5. System Architecture

The system is designed using a layered architecture, where each layer is responsible for a specific part of the automation and processing pipeline. The main components include the frontend web application, backend API services, the AI processing module, and the UiPath automation layer, all working together to transform raw input files into structured knowledge base articles.

5.1. Frontend Web Application

The frontend is developed using React with TypeScript and Vite, styled using Tailwind CSS. It provides a user-friendly interface for interacting with the knowledge base system. The application includes a login page secured with JWT authentication, a dashboard for viewing articles, an upload interface for submitting new files, and dedicated pages for creating, editing, and viewing article details.

The dashboard displays articles in a card-based layout where each card shows the article title, a short summary, and a status indicator. The status reflects whether the article is still being processed, has been successfully completed, or has failed during processing. Users can click on any article card to view full details, edit content, or delete the record.

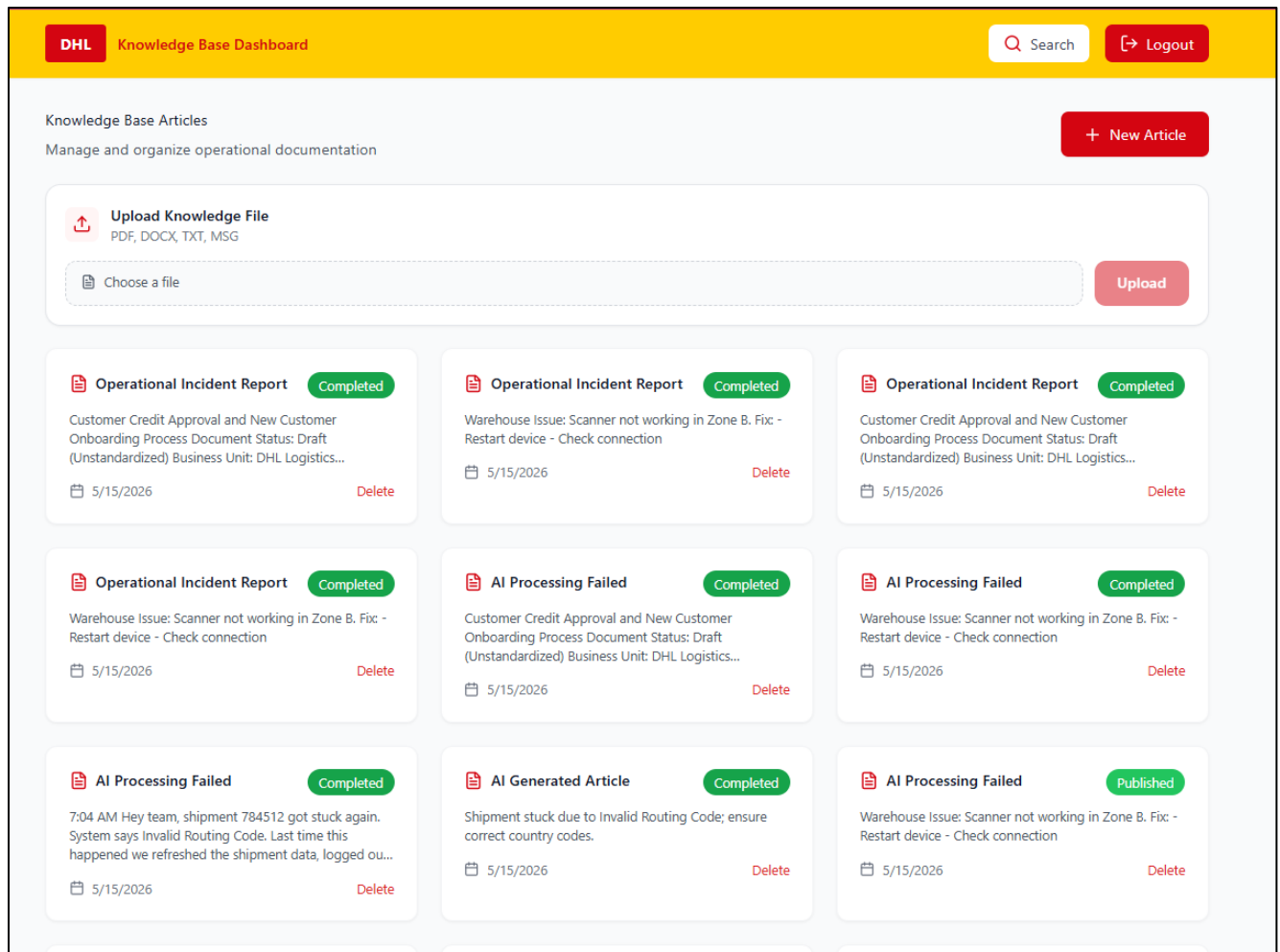


Figure1: Dashboard page after UiPath upload from google drive.

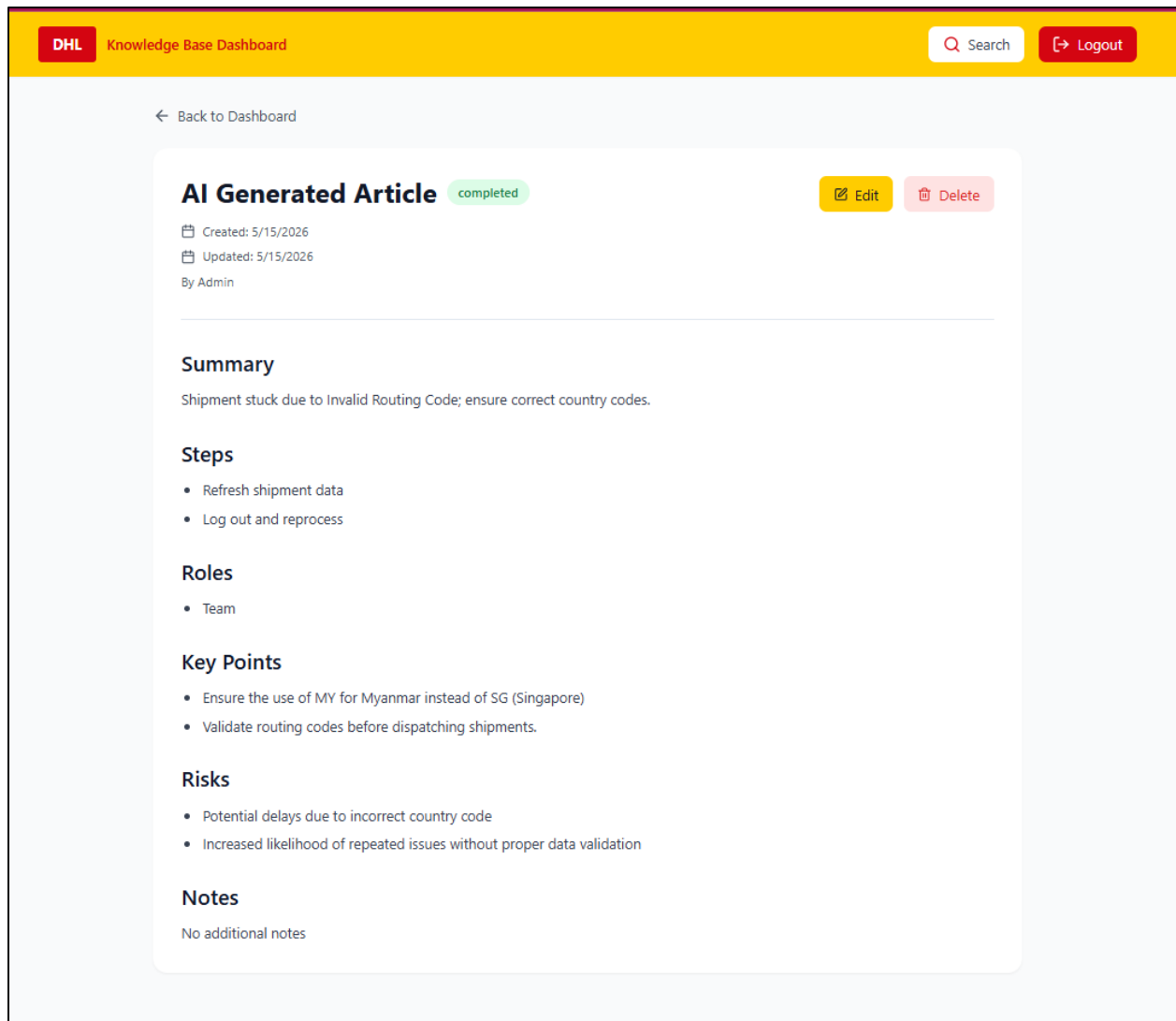


Figure2: details page after Ai summary.

5.2.Backend API Server

The backend is built using Node.js and Express with TypeScript and is connected to a MySQL database managed through XAMPP. It is responsible for handling authentication, file uploads, data processing, and communication with the AI model.

The system includes several core modules. The authentication module manages user login using JWT tokens. The article controller handles CRUD operations and manages the processing pipeline. The file extractor module supports multiple file formats including PDF, DOCX, TXT, and images using OCR techniques. A text cleaning module is used to normalize extracted content before it is sent to the AI model. The LLM service module communicates with the Ollama API to generate structured JSON output.

The backend exposes several RESTful endpoints including login, article retrieval, article upload, and update or deletion of existing articles.

5.3.RPA Layer (UiPath)

The UiPath automation component serves as the trigger for the knowledge base automation pipeline by performing file retrieval and automated communication with the backend API. The workflow is divided into several phases, beginning with authentication, followed by cloud file extraction, and ending with automated payload forwarding.

In the first phase, UiPath performs a background authentication process by sending an HTTP POST request to the `/api/auth/login` endpoint. This request sends login credentials through the request body and captures the response returned by the backend server. The returned response is then processed using the Deserialize JSON activity, which parses the raw response content into a structured JSON object. After deserialization, an Assign activity extracts the JWT token using `jsonObject("token").ToString` and stores it in a global variable named `myToken`. This token is later used to authorize subsequent API requests.

The second phase focuses on cloud file extraction. UiPath uses the Google Workspace Application Scope activity to establish a secure connection to Google Drive through a configured JSON KeyPath credential file. Once connected, the system enters a For Each File or Folder loop that targets a specific Google Drive folder ID. Within this loop, UiPath processes each available file individually.

For every file detected, the Download File activity saves the document into the local processing directory at C:\RPA_Input. A Log Message activity is also used to print the current file name (CurrentItem.Name) to the console, allowing execution tracking and debugging during runtime.

In the final phase, UiPath performs automated payload forwarding by sending another HTTP POST request to the /api/articles/upload endpoint. This request attaches the previously generated JWT token in the Authorization header using the Bearer format (Bearer + myToken). The downloaded file is then streamed to the backend API as a multipart form-data payload.

After the upload is completed, the backend system continues the processing pipeline by extracting file content, cleaning text, sending it to the local AI model, and storing the structured knowledge article in the database. Once all files in the loop have been processed, the UiPath sequence ends successfully.

This UiPath workflow enables end-to-end automation by integrating secure backend authentication, cloud-based file retrieval, automated file downloading, and seamless forwarding of raw input files into the knowledge base processing system.

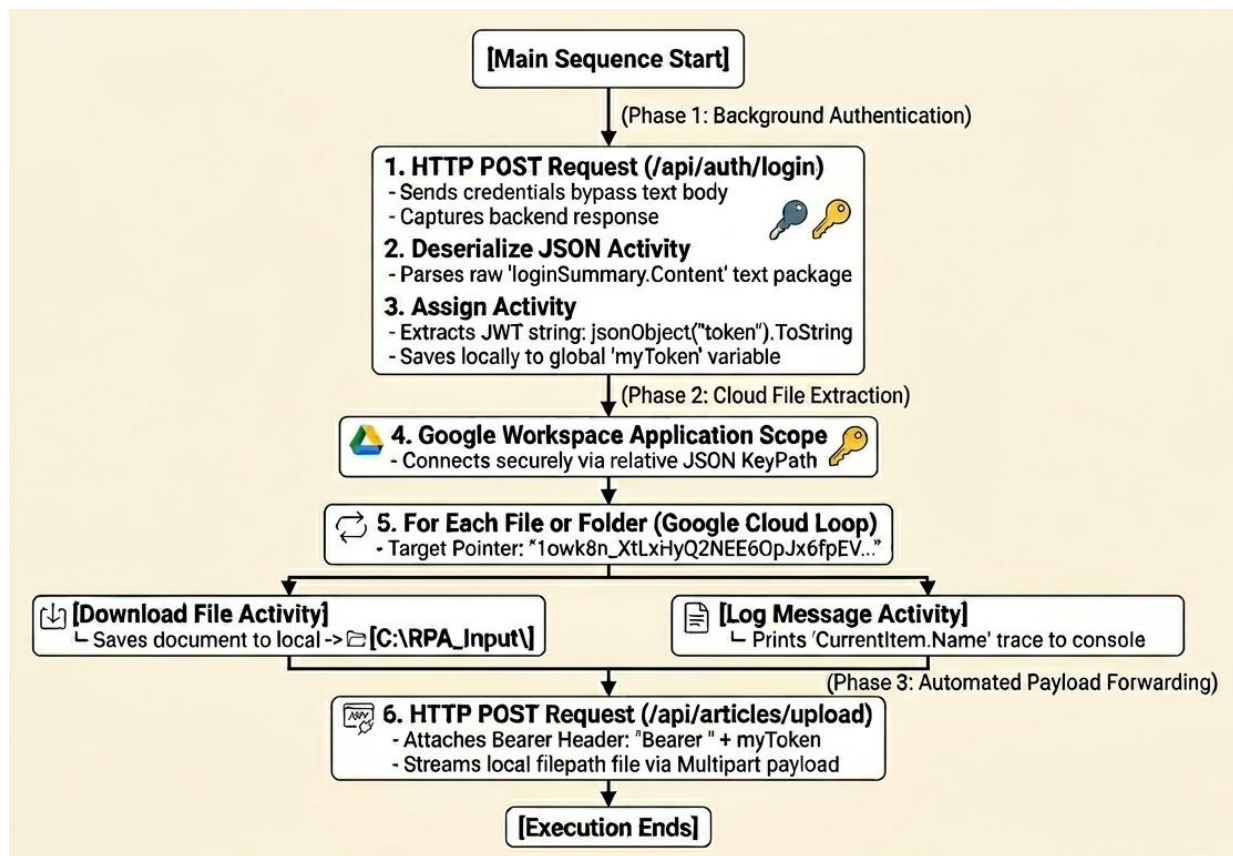


Figure2: UiPath workflow

6. AI Processing Layer

The AI processing layer is powered by a locally hosted Large Language Model (LLM) using Ollama, with models such as Phi-3. Its main role is to analyze cleaned text extracted from uploaded files and transform unstructured operational data into a structured knowledge base format.

After receiving the processed text, the AI identifies important information and organizes it into standardized fields such as title, summary, procedural steps, key points, risks, and notes. This allows raw and inconsistent data to be automatically converted into clear, structured knowledge articles that can be stored, searched, and managed within the system.

7. Database Design

The system uses a single main table called articles in the MySQL database. This table stores both raw and structured data generated during the processing pipeline. It includes fields for a unique identifier, title, full JSON content, summary, structured steps, roles, key points, risks, and notes. It also includes a status field that tracks the processing stage of each article, as well as timestamps for creation and last update.

8. API and Data Flow

The system follows a structured API flow to process uploaded files from either UiPath automation or manual user uploads. When a file is sent to the backend through the upload endpoint using a POST request, the system receives the file and creates an initial database record for tracking.

The uploaded file is then processed through text extraction and cleaning modules, which convert the raw content into plain text and prepare it for analysis. This cleaned text is sent to the integrated Large Language Model (LLM), which restructures the unstructured input into a formatted knowledge base article by generating a title, summary, steps, key points, risks, and notes.

Once the AI response is returned, the backend validates the output, stores the structured article in the database, and updates its status to completed, making it available for viewing and management on the dashboard. This ensures that both automated and manual uploads follow the same AI-driven processing pipeline.

9. User Interface Design

The dashboard interface presents all processed articles in a clean and responsive layout. Each article is displayed as a card containing its title, summary, and current processing status. The status is visually represented using color indicators to distinguish between processing, completed, and failed states. Users can interact with each card to view full details or perform edit and delete operations.

The upload interface is intentionally simple, allowing users to submit files directly into the system. Once uploaded, the file is automatically processed through the RPA and AI pipeline without additional user input. The edit interface allows modification of key fields such as title, summary, notes, and status, ensuring flexibility in managing knowledge entries.

10.RPA Workflow Explanation

The UiPath workflow automates the ingestion of raw files into the knowledge base system. It begins by connecting to a designated Google Drive folder using secure authentication and scanning for files available for processing. Matching files are automatically downloaded into a local staging directory, where they are prepared for transfer into the backend system.

UiPath then processes each file sequentially in a loop, sending them to the backend upload API through multipart form-data HTTP requests. During execution, the workflow generates log messages to track file processing progress and debugging information. This automated approach simulates a real-world enterprise ingestion system where operational files are captured, transferred, and processed continuously without manual intervention.

11.Security Implementation

Security in the system is implemented using JWT-based authentication. All protected routes require a valid token before access is granted. The backend validates these tokens through middleware before allowing access to sensitive endpoints. This ensures that only authenticated users can interact with the system and modify data.

12.System Advantages

The system provides a fully automated end-to-end pipeline that significantly reduces manual effort in knowledge management. It supports multiple file formats and integrates AI to enhance the quality and consistency of structured outputs. The real-time dashboard ensures that users always have access to up-to-date processed information, allowing efficient tracking and management of generated knowledge articles. Overall, the architecture is scalable and designed to replicate enterprise-level document processing workflows commonly used in modern logistics environments such as DHL.

13.Limitations

Despite its advantages, the system is dependent on the performance of the local language model, which may introduce variability in processing time. Large files may take longer to process, and OCR accuracy is dependent on the quality of input images. Additionally, UiPath requires a properly configured local environment to function correctly.

14.Future Improvements

Future enhancements to the system may include integration with cloud-based large language models to improve accuracy and speed. A message queue system such as Redis or RabbitMQ could be introduced to manage high-volume processing more efficiently. Additional features such as advanced search, tagging, and version control could further enhance usability and scalability.

15.Conclusion

The DHL Knowledge Base Automation System successfully demonstrates an end-to-end automation solution that integrates RPA, backend services, and artificial intelligence. It transforms unstructured operational inputs into structured and searchable knowledge articles. This system highlights the potential of combining robotic process automation with AI technologies to improve efficiency, consistency, and scalability in enterprise knowledge management environments.

Appendix

link to video:

<https://drive.google.com/drive/folders/1GEwTjUBgQfltrpaAxoPb4G8DkEeZ0FyK?usp=sharing>